

SHETH LUJ COLLEGE

CYBER SECURITY PRACTICAL - 4

Aim:-Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE:

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CS_Practical4.py - D:/daniyal/CS_Practical4.py (3.14.6)
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import hashlib
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes

# Generate RSA private key
private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

# Get public key
public_key = private_key.public_key()

# Take message from user
message = input("Enter message: ")

# Convert message into bytes
message_bytes = message.encode()

# Create digital signature using private key
signature = private_key.sign(
    message_bytes,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nOriginal Message:", message)
print("Digital Signature:", signature.hex())

# Verify the digital signature
try:
    public_key.verify(
        signature,
        message_bytes,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

# Test message integrity
modified_message = input("\nEnter modified message to test: ")

try:
    public_key.verify(
        signature,
        modified_message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

# Test message integrity
modified_message = input("\nEnter modified message to test: ")

try:
    public_key.verify(
        signature,
        modified_message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

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8/10/2026
```

```
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# Convert message into bytes
message_bytes = message.encode()

# Create digital signature using private key
signature = private_key.sign(
    message_bytes,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nOriginal Message:", message)
print("Digital Signature:", signature.hex())

# Verify the digital signature
try:
    public_key.verify(
        signature,
        message_bytes,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except Exception:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

# Test message integrity
modified_message = input("\nEnter modified message to test: ")

try:
    public_key.verify(
        signature,
        modified_message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

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```

OUTPUT:-

TANVI KHAIR
T088

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```
Python 3.14.6 (tags/v3.14.6:c63aee6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/daniyal/CS_Practical4.py =====
Enter message: Tanvi's roll no is 87

Original Message: Tanvi's roll no is 87
Digital Signature: 5f7403670821e6b3ff2209add5ae9459cc325e33a1651f7528b7a7620bf203d778e824fe2c39d161fc73066cd5f4b7b6943bcb1ed50812c5de75d3ff095f735866c5bdfb0cd4dd62db3f7c7b15add1692eeab6d11ff4df6bf8b01a3417c758e773f11ccce7dca024fef4e58846fde5cd64861bc58934c2dc538da9d57e0e984f694c4c39e2d00a0fb046f7d85567f271c004e446ef8a837debeefc035fcd1bab2228a95b6cc66d6985d124fb2be79355dcca4f4e79cc4d916751243562818758fc37b7d8ccc80abb132e84d2196e3595bee6a3c4e4cd53155be63c966d39cd146258df934997ca2ba2a7daf68552601cdec138dccc3be137f56d66c2389d

Signature Verification: SUCCESS
Message is authentic and has not been modified.

Enter modified message to test: Tanvi's roll no is 87
Modified Message Verification: SUCCESS

>>>
===== RESTART: D:/daniyal/CS_Practical4.py =====
Enter message: I am online

Original Message: I am online
Digital Signature: c47f28aa1c8880985776ed9eab1f0d715ee1a55bd5982f982e843d2352a31308c8bceff64d17b072db30116da36770919c1842f21ca163a0446b56e981e4883783f564b57253f9d2c7259fa851bd830b889a49e39373acf18e87db9e61155c6af0234a5a7e153edcf9c8ee82353bbd18fa8cb06334e256635fc5d6d35cfd13d8c7012ea1bf8b0c3975e10867efb2b456c5853ae54351d29f11fb5149f30b1839e88213c9f48c319d427cc841f6877d02ea189c991cd6557ad6b0e92cdac88bccdd52a5225cc0bdd3e28fc83b7e1923e0023d307f4b6416b689fffd5f997dc593645858ab6e000feb969e02337918ffe52735f1436c2c6f983b843826128c

Signature Verification: SUCCESS
Message is authentic and has not been modified.

Enter modified message to test: I am offline
Modified Message Verification: FAILED
The message has been changed, so authenticity/integrity is not valid.

>>>
```